

PRINCE KUMAR

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Mohali, Punjab, India

Professional Experience

3+ Years Experience

Software Engineer (Team Lead)

Jun 2023 – Present

Sensation Software Solutions Pvt. Ltd.

Mohali, Punjab

- Led development of **TraceVenue** multi-tenant B2B2C venue booking platform using **MERN stack** with **Socket.IO** for real-time booking synchronization, automated proposal workflows, and **JWT authentication** with role-based access control across customer, venue admin, and super-admin roles, scaling to handle **10,000+ venue bookings**; deployed on **AWS EC2** with **Docker** containerization and **S3** for file storage
- Architected scalable **College ERP (CERP)** frontend platform using **React 19** and **TypeScript**, supporting **17+ academic and administrative modules** with **1,300+ component files**; led team of **4–6 developers** defining frontend architecture, component contracts, and performance standards
- Designed **TailUX**, a custom design system on **TailwindCSS 4**, reducing UI development time by **40%**; built intelligent **Exam Seating Planner** with visual grid-based allocation, drag-and-drop assignment, and **PDF/Excel exports**
- Engineered **SensationCRM (LeadNest)** platform using **React 19**, **Vite**, and **TailwindCSS** with lead pipeline management, drag-and-drop form builder, role-based access control, and **TanStack Table** for handling **10,000+ records** with Excel/PDF export and fuzzy search capabilities
- Built **AppyPay** digital payments platform with responsive merchant dashboard using **Material-UI** and reusable compound components; implemented centralized **Redux Toolkit** state management for merchant onboarding, KYC verification, and payment method integration, reducing code duplication and onboarding time by **40%**
- Developed and shipped **3 enterprise GSSC systems** (Accommodation, Vehicle, and Complaint Management), reducing manual operational effort by **50%** and scheduling conflicts by **40%**; built interactive analytics dashboards for **Supra.tools** using **Next.js** and **Chart.js**, visualizing complex feature matrices and reducing stakeholder analysis time by **40%**
- Integrated real-time communication features: **Twilio Voice SDK** for calls, **WhatsApp messaging API**, Firebase notifications, **SignalR** live updates, **ApexCharts** dashboards, multi-language support (**i18n**), and **Sentry** error monitoring achieving **99.9% uptime**
- Implemented **role-based access control (RBAC)**, **JWT authentication**, and secure API endpoints handling **50,000+ API requests daily** with optimized database indexing and **Redis caching** strategies

Software Engineer Intern

Jan 2023 – Jun 2023

Sensation Software Solutions Pvt. Ltd.

Mohali, Punjab

- Built RESTful APIs and React components for **TraceMenu** QR-based restaurant management system, automating order processing and improving operational efficiency by **30%** with real-time kitchen display integration deployed to **50+ restaurants**
- Implemented **Redux store** for cart management with intelligent item aggregation—automatically incremented quantity for duplicate items and maintained cart persistence across page refreshes, reducing redundant API calls
- Utilized **MongoDB transactions** to ensure data consistency across orders, payments, and inventory updates; implemented automatic rollback mechanism on payment failures to maintain database integrity
- Optimized search functionality with debouncing techniques reducing API calls by **60%**; implemented real-time updates via **Socket.IO**, pagination, and email notifications using **Nodemailer**
- Integrated **Cloudinary** for image optimization and CDN delivery achieving **25% reduction** in system response time and improved page load performance

Technical Skills

Languages & Frameworks: JavaScript, TypeScript, Python, React 19, Next.js, React Native, Node.js, Express.js, Fastify
Frontend Technologies: Redux, Zustand, Vite, Tailwind CSS, Material-UI, Bootstrap, Radix UI
Web3 Basics: Web3.js, Ethers.js, Wallet Integration, MetaMask, Smart Contract Interaction
Backend & Databases: MongoDB, PostgreSQL, Prisma, Redis, RabbitMQ, WebRTC, Socket.io, RESTful APIs
Cloud & DevOps: AWS (EC2, ECS, S3, IAM, CloudWatch), Docker, Jenkins, CI/CD, Auto Scaling, Kubernetes
Tools & Practices: Git, GitHub, Jira, Agile/Scrum, Linux, Version Control

Education

CT University

Ludhiana, Punjab

Bachelor of Technology in Computer Science Engineering

2020 – 2024

Key Projects

Virtual Focus Room (Web + Mobile) | *React 19, React Native, WebRTC, Socket.io*

- Built full-stack real-time virtual co-working platform with **P2P video/audio via WebRTC**, screen sharing, collaborative whiteboard, and chat supporting concurrent sessions with tiered permissions (guest/free/premium)
- Implemented custom **Socket.io signaling server** and complete **WebRTC flow** (offer/answer/ICE candidates) using STUN servers; utilized **RTCTrtpSender.replaceTrack** for seamless screen-share toggling without peer renegotiation
- Developed cross-platform experience with **React Native mobile app** and responsive web client; designed VideoGrid component with pinning, spotlight mode, and productivity tools (Pomodoro timers, todos, ambient sounds)
- [🔗 Live Web](#) | [🔗 GitHub](#)

AlgoForgeStudios – Enterprise LMS | *NestJS, Next.js, TypeScript, MongoDB, AWS S3, Razorpay* [🔗 GitHub](#)

- Architected a full-stack LMS using **NestJS** and **Next.js** with **TypeScript**; implemented modular backend with course creation workflows, **RBAC** (Admin, Instructor, Student), **JWT authentication**, and **Mongoose** schemas
- Built secure media pipeline with **AWS S3 presigned URLs** and **CloudFront CDN**, achieving **60% reduction in server load**; handled adaptive delivery for video, audio, and PDF content under heavy media workloads
- Integrated **Razorpay** for end-to-end enrollment and payment flows; implemented real-time progress tracking and analytics dashboards for student and admin roles
- Developed responsive **Next.js** frontend with curriculum builder, assignment tracking, grading interface, and role-specific dashboards; optimized performance with **lazy loading** and **adaptive video streaming**

DrawIO – Collaborative Whiteboard | *Next.js 14, TypeScript, ConvexDB, React Flow* [🔗 Live](#) | [🔗 GitHub](#)

- Built real-time collaborative infinite canvas with **ConvexDB** for instant synchronization, **CRDT-based conflict resolution**, live cursor tracking, and efficient viewport culling for rendering **1,000+ elements**
- Implemented drawing tools (pen, shapes, text), layer management, undo/redo with command pattern, and multi-user presence indicators supporting **50+ concurrent collaborators**

ErgoSmart – AI Posture Coach | *React 18, TensorFlow.js, MediaPipe BlazePose, WebGL* [🔗 Live](#) | [🔗 GitHub](#)

- Built client-side AI posture correction app using **TensorFlow.js** and **MediaPipe BlazePose** achieving **30fps real-time inference** with 33-point body landmark detection and zero backend dependency
- Implemented WebGL acceleration, angle calculation algorithms for posture analysis, and real-time feedback system with audio/visual alerts improving user posture by **35%** in usability testing

VitalWatch – AI Exercise Correction Platform | *React, TensorFlow.js, Pose Estimation* [🔗 GitHub](#)

- Built **VitalWatch** AI-powered exercise correction platform using **TensorFlow.js** with real-time **pose estimation** and skeletal tracking, achieving **100ms inference latency** for instant form feedback with admin interface for exercise configuration

DevOps & Cloud Expertise

AWS Cloud Fundamentals & Infrastructure [🔗 GitHub](#)

- Deployed and managed **Node.js applications** on **AWS ECS (Fargate)** using containerized workloads; built and pushed **Docker images** to **Amazon ECR** with secure ECS service configuration and task definitions
- Automated application deployment using **CI/CD pipelines with Jenkins**, reducing manual deployment effort; configured **AWS EC2 instances** for application hosting with environment setup and process management
- Utilized **Auto Scaling Groups (ASG)** to handle traffic spikes and improve application availability; implemented **AWS IAM** roles and policies for secure service-to-service communication with least-privilege access
- Integrated **Amazon CloudWatch Logs** for centralized logging and monitoring of ECS tasks and services; utilized **Amazon S3** for static asset storage, backup solutions, and configuration management
- Worked with **Docker** to containerize Node.js applications and manage multi-environment builds; familiar with **Kubernetes** concepts including Pods, Deployments, Services, and container orchestration
- Followed **Agile/Scrum methodology** using **Jira** for sprint planning, task tracking, issue management, and collaborative development workflows